

2026 - 0



Carbon Emissions Report 2025

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Introduction

Kumaraguru Institutions is a multi-campus educational institution in Tamil Nadu, South India, offering undergraduate, postgraduate, and research programs in Engineering, Technology, Management, Agriculture, Liberal Arts, and Science. Guided by its motto, "Character is Life," the institution is committed to academic excellence, holistic development, and nurturing responsible individuals with integrity and purpose. Through experiential learning, innovative research, and strong community engagement, Kumaraguru fosters a culture of inquiry and social responsibility, empowering students to address real-world challenges and contribute meaningfully to society.



Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance, resource efficiency, and responsible environmental stewardship.

The program engages students, faculty, and staff in various projects and

Introduction

Kumaraguru Institutions recognize the imperative to address climate change and integrate environmental sustainability into campus operations. This report encapsulates our Greenhouse Gas (GHG) inventory, providing a transparent account of our carbon footprint and resource utilization.

The scope of this report covers all major operational aspects, including energy consumption, transportation, and direct emissions from stationary sources. By quantifying these impacts, we lay the groundwork for targeted reduction strategies and continuous improvement.

Strategic Objectives

- Establish a robust baseline for GHG emissions and energy consumption.
- Identify key emission hotspots to prioritize mitigation interventions.
- Monitor the effectiveness of ongoing sustainability initiatives, such as renewable energy integration.

- Foster a culture of environmental stewardship among students, faculty, and staff.

Highlights



EMISSION PROFILE

Total GHG Emissions

65.69 tCO₂e

Emissions (DG + Transport) **30.6** tCO₂e

Emissions (Electricity) **35.1** tCO₂e

Emissions (LPG - Canteen) **0.00** tCO₂e



ENERGY INDICATORS

Total Energy Consumption

129494 kWh

Grid Energy Consumption **48308** kWh

Renewable Energy Consumption **81186** kWh

Renewable Energy Share **62.7** %

Transport Indicators

Fleet Fuel Consumption Overview



TRANSPORT INDICATORS (FLEET)

Total Fuel Consumption

9586 Litres

Total Petrol Consumption **585.66** Litres

Total Diesel Consumption **9000** Litres

Total Fleet Emissions **25.7** tCO₂e

Climate Action

Sustainability Mitigation Efforts



CLIMATE ACTION INDICATORS

Avoided Emissions

59.0 tCO₂e

Per Capita Emissions

0.000 tCO₂e/person

Executive Summary

High-level overview of carbon performance for Carbon Emissions Report
- Jan 2026.

Performance Overview

During this reporting period, the campus generated 65.69 tCO₂e of gross carbon emissions. Through our renewable energy initiatives, we successfully avoided 59.02 tCO₂e, resulting in a net carbon impact of 6.67 tCO₂e. We remain committed to our long-term decarbonization pathway.



CARBON STORY

Gross GHG Emissions

65.7 tCO₂e

Net GHG Emissions

6.7 tCO₂e

Avoided Emissions

59.0 tCO₂e



STRATEGIC RECOMMENDATIONS

- Accelerate the transition of internal campus mobility to Electric Vehicles (EVs) to reduce Scope 1 transport emissions.
- Expand rooftop solar capacity to push the renewable energy mix beyond the current baseline.

- Implement automated BMS (Building Management Systems) for high-load commercial blocks to optimize HVAC consumption.

Scope 1 Story

Direct Emissions Overview

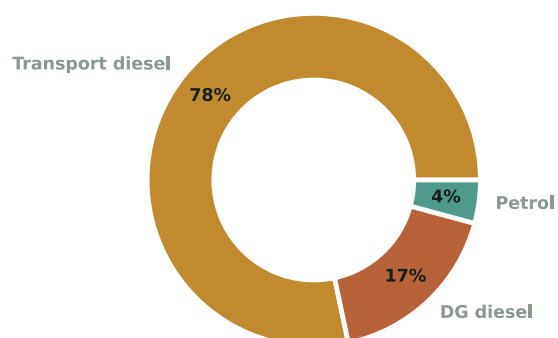
Scope 1 emissions generated 30.57 tCO₂e. Transport remains the dominant emission source in this category. Overall Scope 1 emissions decreased by 5.7% compared to the previous period.

Transport Deep Dive

Transport emissions decreased by 8.6%. Continued fleet optimization is contributing to this reduction.

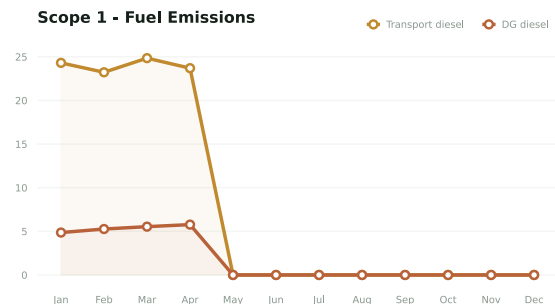
Scope 1 Composition

Scope 1 Breakdown



Emissions Trend

Scope 1 - Fuel Emissions



Transport Emissions (Scope 1)

25.7 tCO₂e

MONTH	PETROL (LITRES)	EMISSIONS (TCO ₂ E)	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	585.7	1.3986	9000	24.3090
FEB	520.0	1.2418	8600	23.2286
MAR	540.0	1.2895	9200	24.8492
APR	508.4	1.2140	8775	23.7002

DG Set Emissions (Scope 1)

4.9 tCO2e

MONTH	DIESEL (LITRES)	EMISSIONS (TCO2E)
JAN	1800.00	4.8618
FEB	1950.00	5.2669
MAR	2050.00	5.5370
APR	2136.83	5.7716

Scope 2 Story

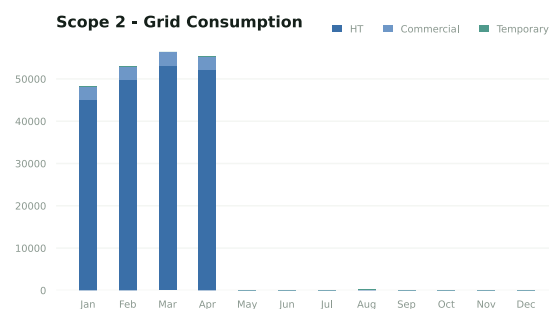
Indirect Emissions Overview

Grid electricity consumption accounted for 35.12 tCO₂e. This represents 53.5% of total gross emissions. Scope 2 emissions fell by 69.1%. Continuous efforts in optimizing HVAC and lighting can further reduce this footprint.

Renewable Impact

Renewable energy integration contributed 81186.00 kWh, offsetting approximately 59.02 tCO₂e of potential emissions. Our current renewable share stands at 62.7%.

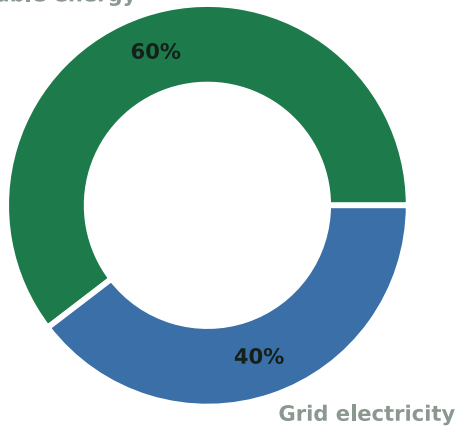
Grid Consumption Profile



Renewable Energy Mix

Renewable Mix

Renewable energy



Electricity Emissions (Scope 2)

35.1 tCO2e

RENEWABLE ENERGY

81186 kWh

GRID ENERGY

48308 kWh

TOTAL CONSUMPTION

129494 kWh

MONTH	GRID HT		GRID COMMERCIAL		GRID TEMPORARY	
	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)
JAN	45208	32.866	3100	2.254	0	0.000
FEB	49822	36.221	3042	2.212	0	0.000
MAR	53200	38.676	3200	2.326	0	0.000
APR	52065	37.851	3185	2.315	0	0.000
AUG	100	0.073	0	0.000	0	0.000

Methodology

Purpose of the Report

This report presents the Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions for the reporting period, compiled in accordance with ISO 14064-1:2018. The objective of this inventory is to measure and disclose the institution's GHG emissions, and provide reliable data to support future emissions reduction strategies.

GHG Quantification Approach

The inventory has been developed using an activity-based quantification approach. Emissions were calculated by multiplying activity data (e.g., fuel consumption, electricity consumption) with appropriate emission factors derived from recognized national and international sources (IPCC, CEA, DEFRA).

Organizational Boundary

The organizational boundary has been defined using the operational control approach. All facilities and operations over which Kumaraguru Institutions has operational control are included. Scope 3 emissions have been assessed at a screening level but are excluded from quantitative reporting due to the unavailability of consolidated data.

Global Warming Potential (GWP)

All greenhouse gases were converted into carbon dioxide equivalent (CO₂e) using Global Warming Potential (GWP) values from the IPCC Sixth Assessment framework.

Conclusion

This Greenhouse Gas (GHG) inventory enhances the understanding of emission sources within campus operations and provides a foundation for tracking environmental performance and identifying opportunities for energy efficiency and emission reduction. The institution intends to progressively improve data quality, expand emission coverage, and integrate climate considerations into operational decision-making.

REPORT PREPARED BY

Microcosm Sustainability Intelligence

A long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance.